

Experiment-2

Verification of the Overfitting Issue Using a Numerical Dataset in Python

✓ AIM

To verify the occurrence of **overfitting** by training a simple model and a high-complexity model on a numerical dataset and comparing their training and testing errors using Python.

✓ ALGORITHM

Algorithm to Verify Overfitting

1. **Start**
2. Generate or load a numerical dataset.
3. Split the dataset into **training** and **testing** sets.
4. Fit a **simple model** (Linear Regression).
5. Predict values for both training and testing sets.
6. Compute **training error** and **testing error** using Mean Squared Error (MSE).
7. Fit a **high-complexity model** (10-degree Polynomial Regression).
8. Predict values for both training and testing sets.
9. Compute **training error** and **testing error** for this model.
10. Compare errors:
 - If **training error** << **testing error**, overfitting is occurring.
11. Display the errors and conclude.
12. **Stop**

✓ INPUT & OUTPUT

```
main.py  [Copy] [Refresh] [Share] [Run] [Clear]

1 # Overfitting verification (fixed)
2 import numpy as np
3
4 # Try to import plotting (optional). If unavailable,
  continue without plotting.
5 try:
6     import matplotlib.pyplot as plt
7     plotting_available = True
8 except Exception:
9     plotting_available = False
10
11 # ----- 1. Create Numerical Dataset -----
12 np.random.seed(0)
13 x = np.linspace(0, 5, 20)
14 y = 2 * x + np.random.randn(20) # linear relation + noise
15
16 # Split into train and test
17 x_train, y_train = x[:15], y[:15]
18 x_test, y_test = x[15:], y[15:]
19
20 # ----- 2. Fit Simple Linear Model (No Overfitting)
  -----
21
22 # Closed-form slope (a) and intercept (b)
```

```
=== LINEAR MODEL (NO OVERFITTING) ===
ERROR!
Training Error: 0.6274139324428416
Testing Error : 0.5476015039721627

=== POLYNOMIAL MODEL (POSSIBLE OVERFITTING) ===
Training Error: 0.20152074678221965
Testing Error : 121432178.49450362

Plotting not available in this environment. MSE values printed above
.

=== Code Execution Successful ===
```

✓ RESULT

The experiment successfully demonstrated the **overfitting issue**.

The **polynomial (high-degree) model** achieved a very low training error but a very high testing error, proving **overfitting**.

- **Linear Model:**
 - Training Error $\approx$ Testing Error $\rightarrow$ **No Overfitting**
- **Polynomial Model:**
 - Training Error **very low**
 - Testing Error **very high** $\rightarrow$ **Overfitted model**

Thus, the performance difference clearly verifies **overfitting** in the high-complexity model.